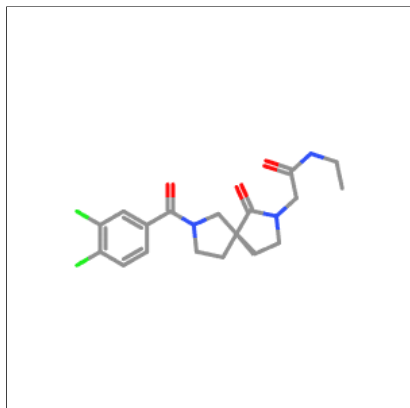
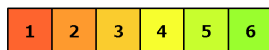


## Oral toxicity prediction results for input compound



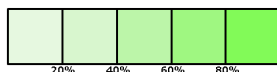
Predicted LD50: 460mg/kg

Predicted Toxicity Class: 4



Average similarity: 71.21%

Prediction accuracy: 69.26%



|                                           |                     |
|-------------------------------------------|---------------------|
| Name                                      | CCNC(=O)CN1CC[C@]2( |
| Molweight                                 | 398.28              |
| Number of hydrogen bond acceptors         | 6                   |
| Number of hydrogen bond donors            | 1                   |
| Number of atoms                           | 26                  |
| Number of bonds                           | 28                  |
| Number of rotatable bonds                 | 6                   |
| Molecular refractivity                    | 107.06              |
| Topological Polar Surface Area            | 69.72               |
| octanol/water partition coefficient(logP) | 2.46                |

## Toxicity Model Report

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| Classification                             | Target                                                                                                | Shorthand     | Prediction | Probability |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------|------------|-------------|
| Organ toxicity                             | <a href="#">Hepatotoxicity</a>                                                                        | dili          | Inactive   | 0.82        |
| Organ toxicity                             | <a href="#">Neurotoxicity</a>                                                                         | neuro         | Active     | 0.74        |
| Organ toxicity                             | <a href="#">Nephrotoxicity</a>                                                                        | nephro        | Inactive   | 0.62        |
| Organ toxicity                             | <a href="#">Respiratory toxicity</a>                                                                  | respi         | Active     | 0.81        |
| Organ toxicity                             | <a href="#">Cardiotoxicity</a>                                                                        | cardio        | Inactive   | 0.85        |
| Toxicity end points                        | <a href="#">Carcinogenicity</a>                                                                       | carcino       | Inactive   | 0.65        |
| Toxicity end points                        | <a href="#">Immunotoxicity</a>                                                                        | immuno        | Inactive   | 0.92        |
| Toxicity end points                        | <a href="#">Mutagenicity</a>                                                                          | mutagen       | Inactive   | 0.70        |
| Toxicity end points                        | <a href="#">Cytotoxicity</a>                                                                          | cyto          | Inactive   | 0.69        |
| Toxicity end points                        | <a href="#">BBB-barrier</a>                                                                           | bbb           | Active     | 0.69        |
| Toxicity end points                        | <a href="#">Ecotoxicity</a>                                                                           | eco           | Active     | 0.55        |
| Toxicity end points                        | <a href="#">Clinical toxicity</a>                                                                     | clinical      | Active     | 0.70        |
| Toxicity end points                        | <a href="#">Nutritional toxicity</a>                                                                  | nutri         | Inactive   | 0.56        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Aryl hydrocarbon Receptor (AhR)</a>                                                       | nr_ahr        | Inactive   | 0.91        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Androgen Receptor (AR)</a>                                                                | nr_ar         | Inactive   | 0.92        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Androgen Receptor Ligand Binding Domain (AR-LBD)</a>                                      | nr_ar_lbd     | Inactive   | 0.99        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Aromatase</a>                                                                             | nr_aromatase  | Inactive   | 0.93        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Estrogen Receptor Alpha (ER)</a>                                                          | nr_er         | Inactive   | 0.84        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Estrogen Receptor Ligand Binding Domain (ER-LBD)</a>                                      | nr_er_lbd     | Inactive   | 0.98        |
| Tox21-Nuclear receptor signalling pathways | <a href="#">Peroxisome Proliferator Activated Receptor Gamma (PPAR-Gamma)</a>                         | nr_ppar_gamma | Inactive   | 0.98        |
| Tox21-Stress response pathways             | <a href="#">Nuclear factor (erythroid-derived 2)-like 2/antioxidant responsive element (nrf2/ARE)</a> | sr_are        | Inactive   | 0.94        |
| Tox21-Stress response pathways             | <a href="#">Heat shock factor response element (HSE)</a>                                              | sr_hse        | Inactive   | 0.94        |
| Tox21-Stress response pathways             | <a href="#">Mitochondrial Membrane Potential (MMP)</a>                                                | sr_mmp        | Inactive   | 0.88        |
| Tox21-Stress response pathways             | <a href="#">Phosphoprotein (Tumor Suppressor) p53</a>                                                 | sr_p53        | Inactive   | 0.90        |
| Tox21-Stress response pathways             | <a href="#">ATPase family AAA domain-containing protein 5 (ATAD5)</a>                                 | sr_atad5      | Inactive   | 0.98        |
| Molecular Initiating Events                | <a href="#">Thyroid hormone receptor alpha (THRα)</a>                                                 | mie_thr_alpha | Inactive   | 0.94        |
| Molecular Initiating Events                | <a href="#">Thyroid hormone receptor beta (THRβ)</a>                                                  | mie_thr_beta  | Inactive   | 0.77        |
| Molecular Initiating Events                | <a href="#">Transthyretin (TTR)</a>                                                                   | mie_ttr       | Inactive   | 0.95        |
| Molecular Initiating Events                | <a href="#">Ryanodine receptor (RYR)</a>                                                              | mie_ryr       | Inactive   | 0.83        |
| Molecular Initiating Events                | <a href="#">GABA receptor (GABAR)</a>                                                                 | mie_gabar     | Inactive   | 0.58        |

| Classification              | Target                                                               | Shorthand  | Prediction | Probability |
|-----------------------------|----------------------------------------------------------------------|------------|------------|-------------|
| Molecular Initiating Events | Glutamate N-methyl-D-aspartate receptor (NMDAR)                      | mie_nmdar  | Inactive   | 0.94        |
| Molecular Initiating Events | alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionate receptor (AMPA) | mie_ampar  | Inactive   | 0.99        |
| Molecular Initiating Events | Kainate receptor (KAR)                                               | mie_kar    | Inactive   | 0.99        |
| Molecular Initiating Events | Achetylcholinesterase (AChE)                                         | mie_ache   | Inactive   | 0.61        |
| Molecular Initiating Events | Constitutive androstane receptor (CAR)                               | mie_car    | Inactive   | 0.99        |
| Molecular Initiating Events | Pregnane Xreceptor (PXR)                                             | mie_pxr    | Active     | 0.53        |
| Molecular Initiating Events | NADH-quinone oxidoreductase (NADHOX)                                 | mie_nadhox | Inactive   | 0.95        |
| Molecular Initiating Events | Voltage gated sodium channel (VGSC)                                  | mie_vgsc   | Inactive   | 0.77        |
| Molecular Initiating Events | Na+/- symporter (NIS)                                                | mie_nis    | Inactive   | 0.96        |
| Metabolism                  | Cytochrome CYP1A2                                                    | CYP1A2     | Inactive   | 0.95        |
| Metabolism                  | Cytochrome CYP2C19                                                   | CYP2C19    | Inactive   | 0.73        |
| Metabolism                  | Cytochrome CYP2C9                                                    | CYP2C9     | Inactive   | 0.58        |
| Metabolism                  | Cytochrome CYP2D6                                                    | CYP2D6     | Active     | 0.52        |
| Metabolism                  | Cytochrome CYP3A4                                                    | CYP3A4     | Inactive   | 0.73        |
| Metabolism                  | Cytochrome CYP2E1                                                    | CYP2E1     | Inactive   | 1.0         |

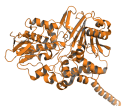
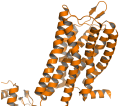
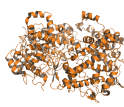
Toxicity targets

Possible binding to toxicity targets is shown below. For more information on the targets, please click on the individual abbreviations.



|                       |                       |                      |                      |                       |                      |                      |                      |                     |                      |                       |                      |                      |                       |                      |                      |
|-----------------------|-----------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|----------------------|---------------------|----------------------|-----------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|
| <a href="#">AA2AR</a> | <a href="#">ADRB2</a> | <a href="#">ANDR</a> | <a href="#">AOFA</a> | <a href="#">CRFR1</a> | <a href="#">DRD3</a> | <a href="#">ESR1</a> | <a href="#">ESR2</a> | <a href="#">GCR</a> | <a href="#">HRH1</a> | <a href="#">NR1I2</a> | <a href="#">OPRK</a> | <a href="#">OPRM</a> | <a href="#">PDE4D</a> | <a href="#">PGH1</a> | <a href="#">PRGR</a> |
|                       |                       |                      |                      |                       |                      |                      |                      |                     |                      |                       |                      |                      |                       |                      |                      |

Details about possible toxicity targets:

|                                                                                     | Toxicity Target              | Avg Pharmacophore Fit | Avg Similarity Known Ligands |
|-------------------------------------------------------------------------------------|------------------------------|-----------------------|------------------------------|
|  | Amine Oxidase A              | 47.63%                | 0%                           |
|  | Opioid Receptor Mu           | 0%                    | 74.49%                       |
|  | Prostaglandin G/H Synthase 1 | 46.84%                | 0%                           |